

CSE 318 Assignment-02 Report

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1 High-Level Description of Each Algorithm

1.1 Randomized Algorithm

We put each vertex in one of two sets based on 50/50 probability, then repeat many trials and take the average or best cut found.

1.2 Greedy Algorithm

We start by picking the heaviest edge and put its endpoints in opposite sets, then for each remaining vertex we add it to the side that immediately gives the largest gain in cut weight.

1.3 Semi-Greedy Algorithm

We compute a score for each unassigned vertex, build a Restricted Candidate List based on a value cutoff μ , pick one at random, and place it to maximize cut weight, repeating until all vertices are assigned.

1.4 Local Search

From an initial cut we consider moving each vertex to the other side, compute the change $\delta(v)$, and perform the best improving move until no positive δ exists.

1.5 GRASP

We combine the semi-greedy construction with local search, repeat this process for a fixed number of iterations, and keep the best solution over all runs.

2 Comparison of Algorithms

We compare performance across all benchmark graphs. Generally, the Greedy heuristic outperforms pure Randomized on small to medium instances, but GRASP achieves the highest cut values overall at the cost of more computation time. Semi-Greedy often finds better cuts than Greedy alone by introducing controlled randomness, while Local Search improves all initial solutions reliably.

3 Plots and Visualizations

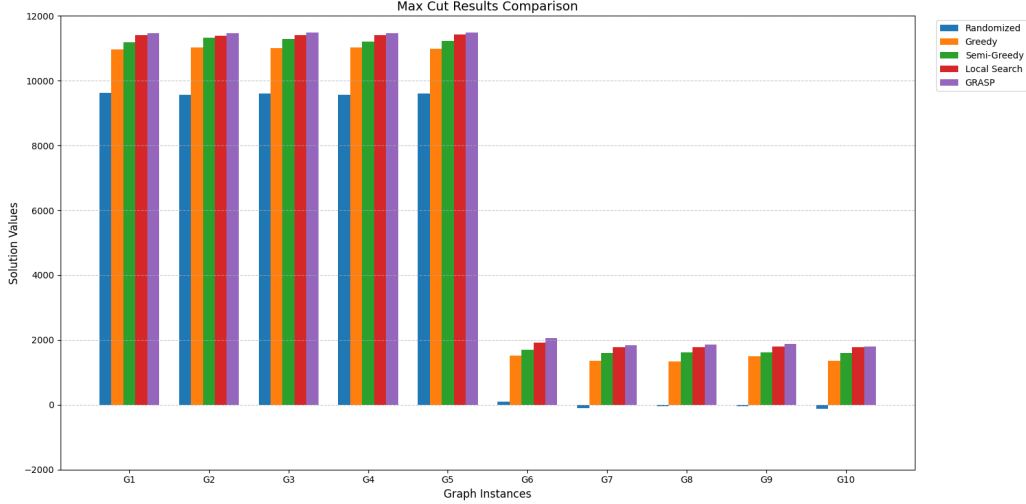


Figure 1: Comparison of cut weights across different graph